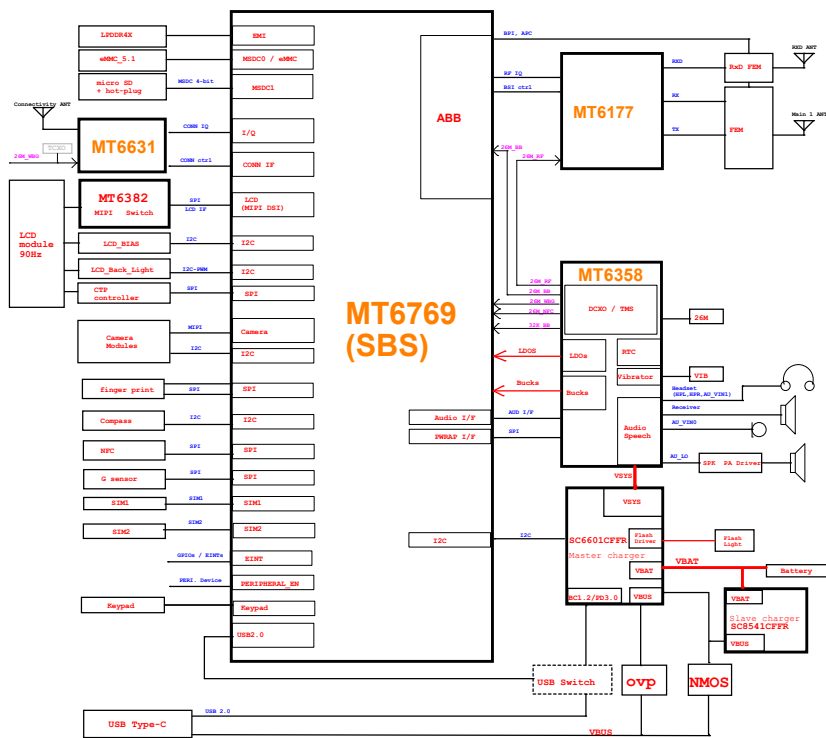




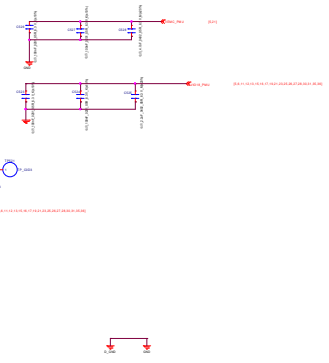
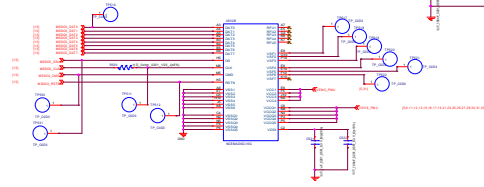
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 01_Block_Diagram		
Size D	Project AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024	Sheet 1 of 36	

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 02_I2C_ID_Overview		
Size C	Project AL7015A_MB_V1	Rev V1
Date: Monday, January 01, 2024		Sheet 2 of 36

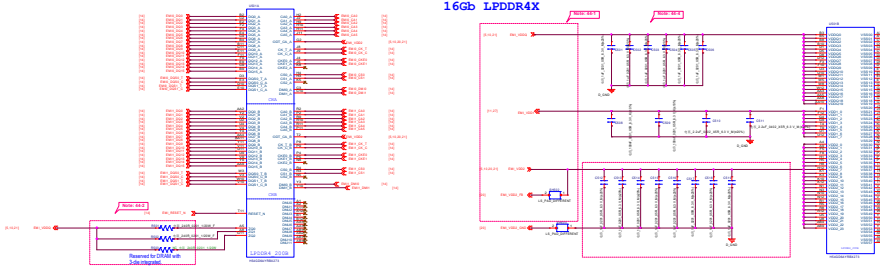
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 03_Change_Notice		
Size C	Project AL7015A_MB_V1	Rev V1
Date: Monday, January 01, 2024		Sheet 3 of 36

华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title 04_CHANGELIST		
Size D	Project AL7015A_MB_V1	Rev V1
Date: Monday, January 01, 2024		Sheet 4 of 36

emmc 5.1

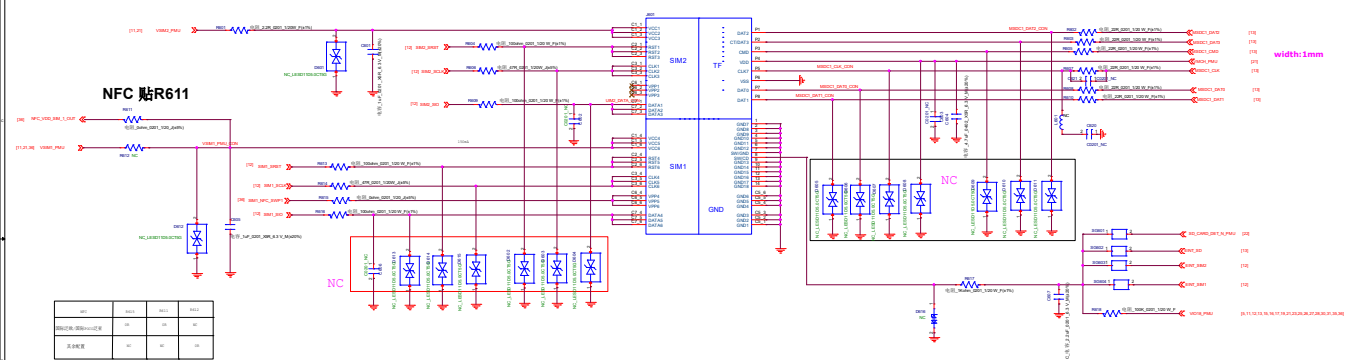


discrete LP4X 16Gb LPDDR4X



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 05_Memory_eMMC_LPDDR4			
Size E	Project	AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024		Sheet 5 of 36	

3选2卡座



华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 06_Memory_SD_SIM_Card

Size
D

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Date: Monday, January 01, 2024

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[illegible][illegible]

The diagram shows a 4-wire RTD circuit. A 5V supply is connected to a 10k pull-up resistor, which is connected to the ACC_P pin. The ACC_P pin is also connected to one terminal of a 250 Ohm RTD. The other terminal of the RTD is connected to the ACC_N pin. The ACC_N pin is connected to ground. The ACC_V pin is connected to ground. The ACC_W pin is connected to a 5V supply. The circuit is labeled 'RTD (250 Ohm) for ACC module'.

[illegible]

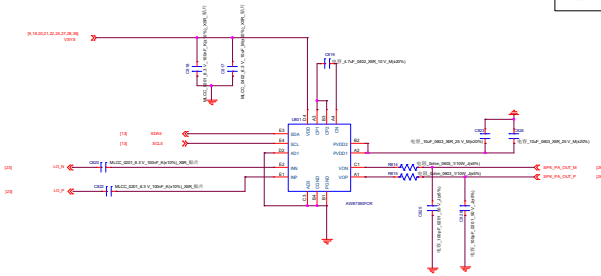
Slave MIC

Title	07_AUDIO_REC_MAIN_MIC_HEADPHON
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Date: Monday, January 01, 2024	Sheet 7 of 36
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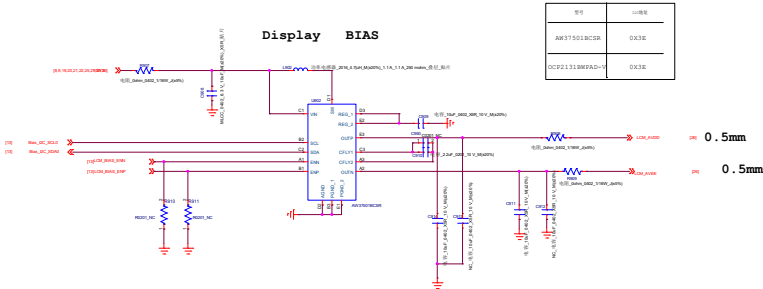
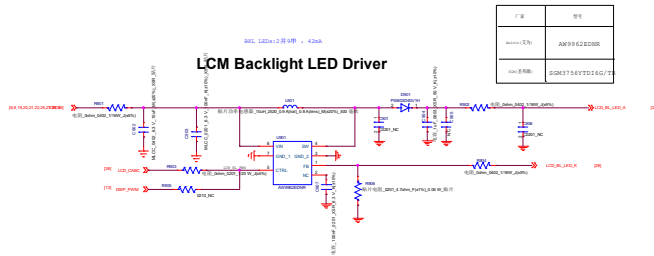
[illegible]

SPEAKER



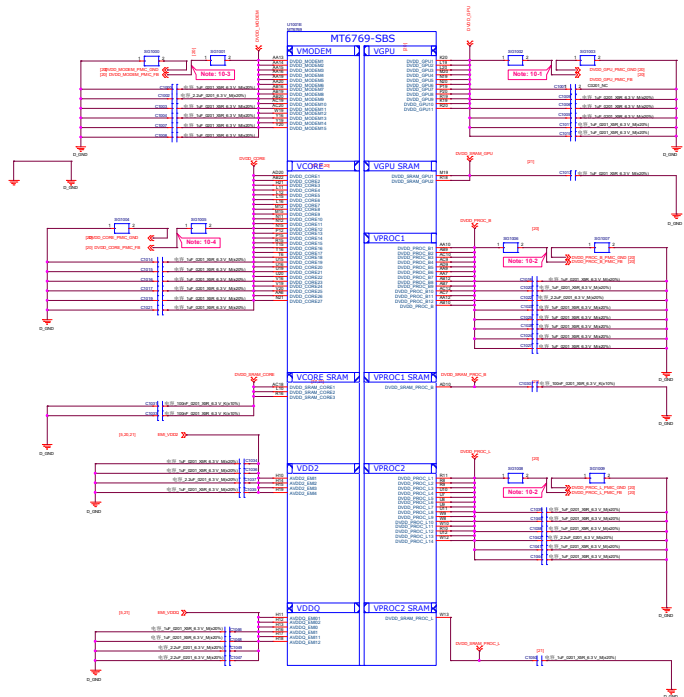
01	06
0007733DFC8	0X58
7F0255980	0034

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 08_AUDIO_SPEAKER_AMP		
Size D	Project AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024		Sheet 8 of 36



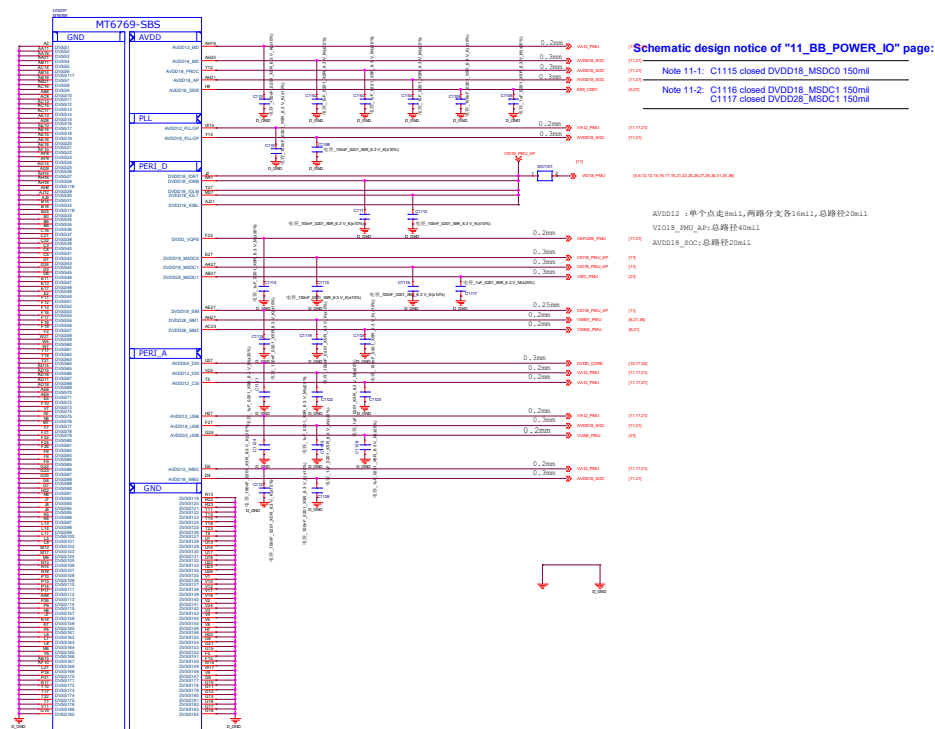
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 09_LCD_CTP_BL		
Size D	Project AL7015A_MB_V1	Rev V1
Date: Monday, January 01, 2024	Sheet 9 of 36	

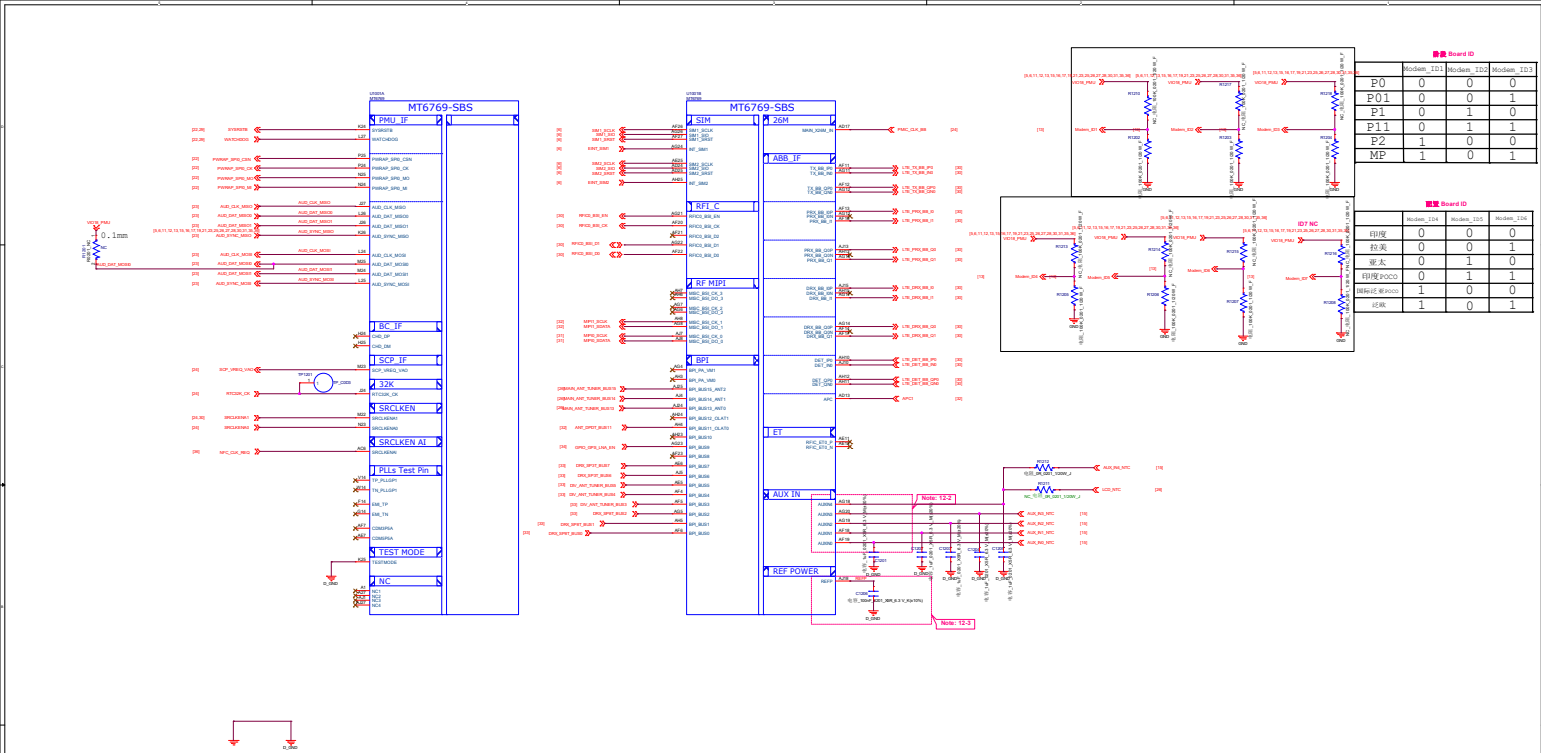
Schematic design notice of "10_BB_POWER_PDN" page:



- Note 10-1: Differential pair of DVDD_GPU_REMOTE_SENSE must be close to BB's ball.
Remote sense trace with GND shielding to PMC (Differential)
- Note 10-2: Differential pair of DVDD_CORE_REMOTE_SENSE must be close to BB's ball.
Remote sense trace with GND shielding to PMC (Differential)
- Note 10-3: Differential pair of DVDD_MODEM_REMOTE_SENSE must be close to BB's ball.
Remote sense trace with GND shielding to PMC (Differential)
- Note 10-4: Differential pair of DVDD_CORE_REMOTE_SENSE must be close to BB's ball.
Remote sense trace with GND shielding to PMC (Differential)

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		10_BB_POWER_PDN	
Size	Project	AL7015A_MB_V1	Rev
D			V1
Date:	Monday, January 01, 2024	Sheet	10 of 36





Schematic design notice of "12_BB_1" page:

Note 12-1: PWRAP_SPI0_CSN and "AUD_DAT_MOSI0" are bootstrap pins to select which interface will be the JTAG pin out

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	JTAG Function
defaultP0	defaultP0	AP_JTAG
HI	LO	MD_JTAG
LO	HI	SPI0 + ENTB
LO	LO	SPI0 + ENTB

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

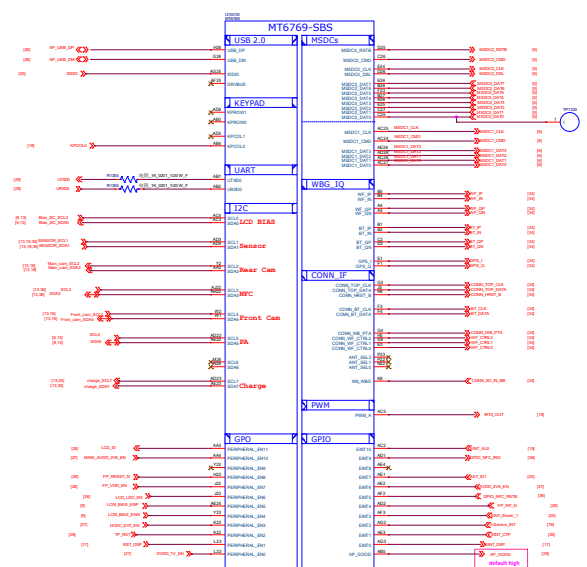
Note 12-3: The de-coupling cap. for REFP (AJ18 ball) have to be placed as close to BB as possible.

Note 12-4: HW pin for DDR type feature in bootstrap (refer to BB HW design Notice)

AUD_SYNC_MISO	AUD_CLK_MISO	CAS_PEND	PMIC 6388 voltage	DDR Type
defaultP0	defaultP0	defaultP0	VORAM7 VORAM2	DDR
No ext. pull	No ext. pull	No ext. pull	1.125V / 0.6V	LPRX aMCP
No ext. pull	No ext. pull	No ext. pull	1.125V / 1.1V	LPX aMCP
No ext. pull	No ext. pull	No ext. pull	1.125V / 1.1V	LPX aMCP

Note 12-5: Charger must have D+/D- pin for charger type USB detection. Charger must have at least 500mA USB current for All charger type.

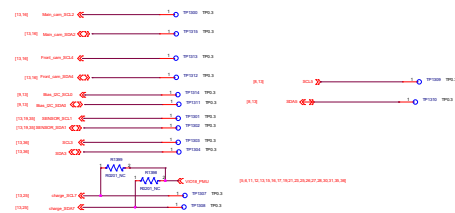
华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title		12_BB1L	
Size	Project	AL7015A_MB_V1	Rev v1
Date:	Monday, January 01, 2024	Sheet	12 of 36



- MT6768/67 DDR trapping table

	MT6768 / MT6767				PMIC MT6768	
	AUG_SYNC_MISO (K16: K20)	AUG_CLK_MISO (K16: K27)	CAIT_P0K3 (K16: K4)		VDRAM1	VDRAM2
EMCP LP4K (254 B)	No ext. pull	No ext. pull	No ext. pull		1.125V	0.6V
Reserved	No ext. pull	12K to VIO18	No ext. pull		0V	1.8V
EMCP LP3 (224 B)	12K to VIO18	No ext. pull	No ext. pull		1.225V	0V
Reserved	12K to VIO18	12K to VIO18	No ext. pull		1.125V	1.8V
Discrete LP4K (200 B)	No ext. pull	No ext. pull	12K to VIO18		1.225V	0.6V
Reserved	No ext. pull	12K to VIO18	12K to VIO18		0V	1.8V
Reserved	12K to VIO18	No ext. pull	12K to VIO18		1.225V	0V
Reserved	12K to VIO18	12K to VIO18	12K to VIO18		1.125V	1.8V

I2C TEST



华勤通讯 Huaqin Telecom Technology Com.,Ltd

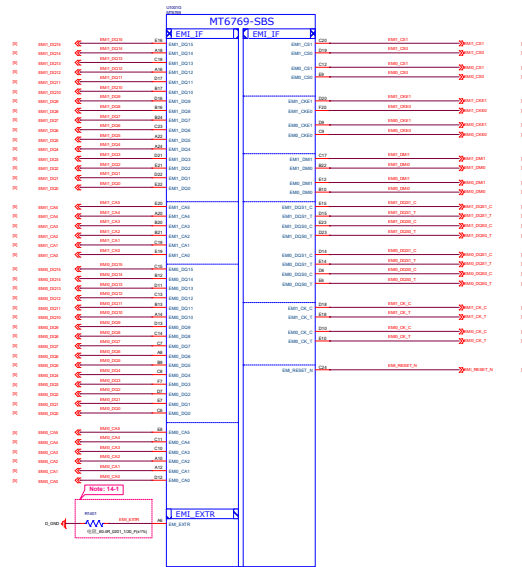
Title	13_BB2
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Size D	Project AL7015A_MB_V1
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Rev
V1

Date: Monday, January 01, 2024

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Schematic design notice of "14_BB_3" page.

Note 14-1: R1401 please select 60.4 ohm (1%) resistor

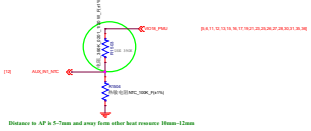
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		14_BB3	
Size	Project	AL7015A_MB_V1	Rev
D			v1
Date:	Monday, January 01, 2024		Sheet 14 of 36

CPU_THERMAL



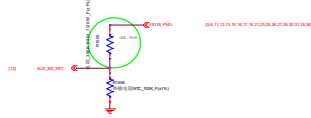
change to NCP03WF104F05RL Murata

4G_PA_THERMAL

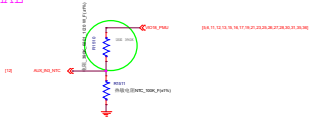


Distance to AP is 5mm and away from other heat sources (Hsiao-Tsann)

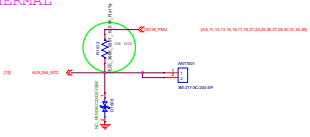
charger



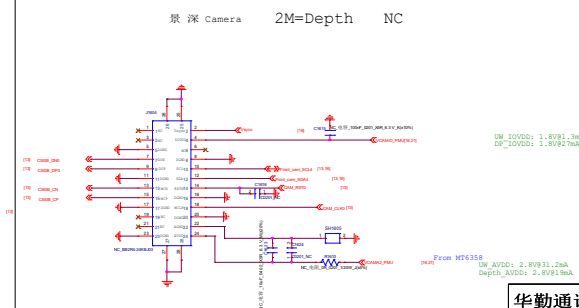
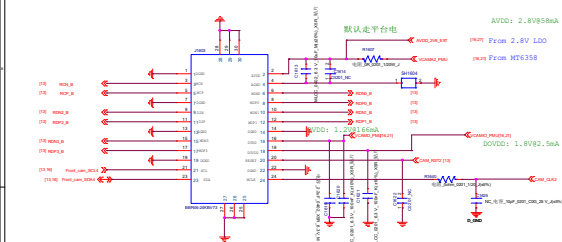
BOARD_THERMAL



FlashLED_THERMAL



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 15_BB_AUXADC_Thermal			
Size D	Project AL7015A_MB_V1		Rev V1
Date: Monday, January 01, 2024		Sheet 15 of 36	

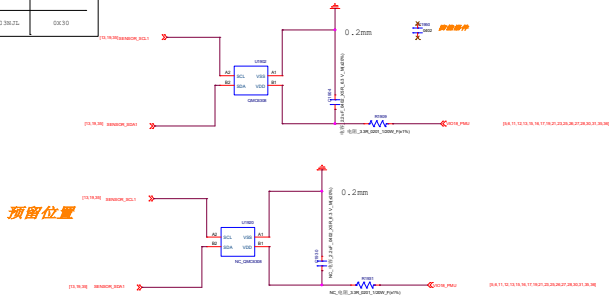


Haqin Telecom Technology Co.,Ltd		
Title 16_CAMERA_INTERFACE		
Size D	Project AL7015A_MB_V1	Rev V1
Date: Monday, January 01, 2024	Sheet 16 of 36	

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 18_KEYPAD_INTERFACE		
Size D	Project AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024	Sheet 18 of 36	

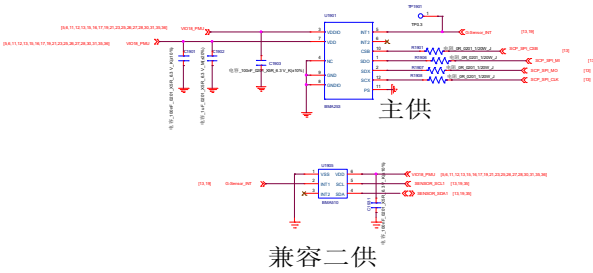
eCOMPASS

型号	封装
QMC6308	DSOC
HMCC360 DSLS	DS30



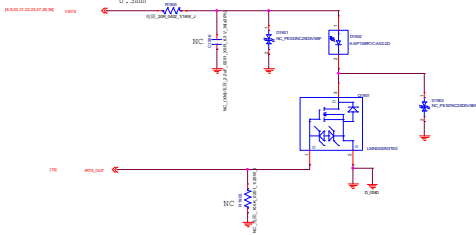
G-Sensor

型号	封装
BM253	DS18
BM251C	DSOF



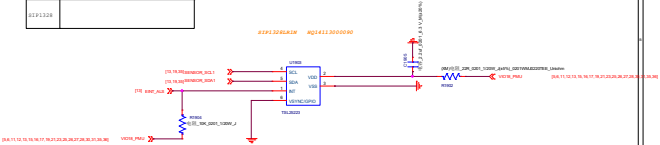
IR LED

型号	封装
ESD6-ESD-4550-B-C	ESD6-ESD-4550-B-C
ESD6-ESD-4550-B-C	ESD6-ESD-4550-B-C

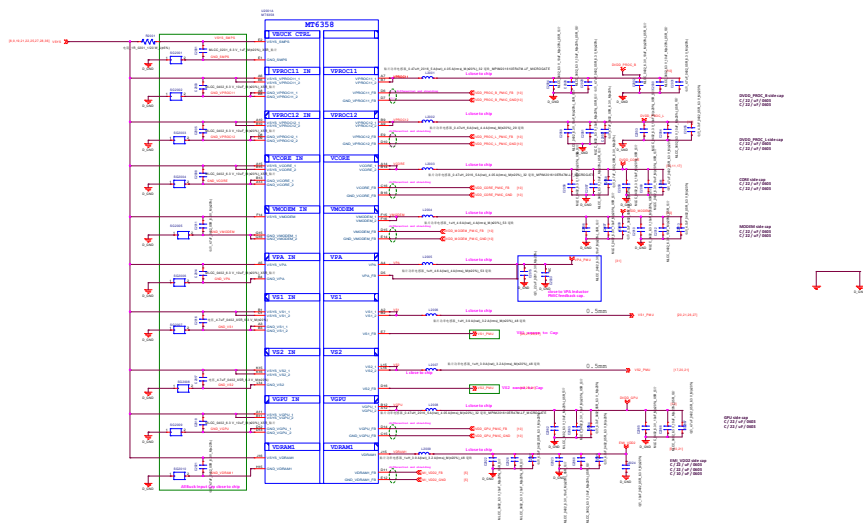


ALS

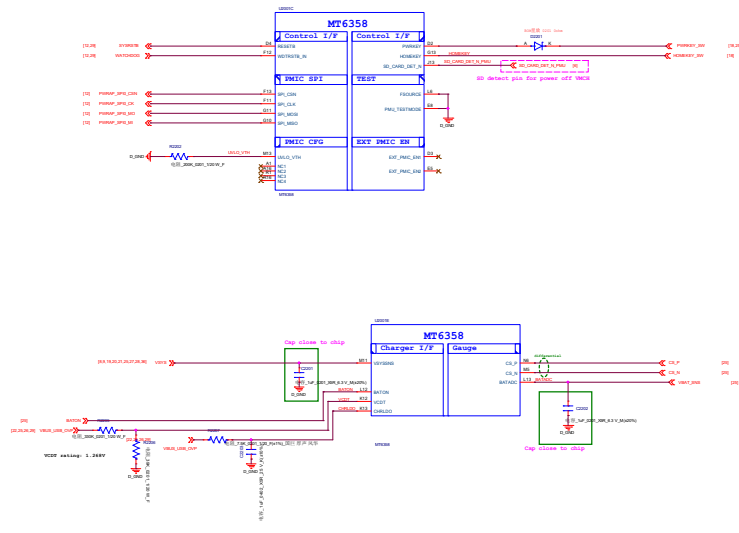
型号	封装
ALS1224	ALS1224



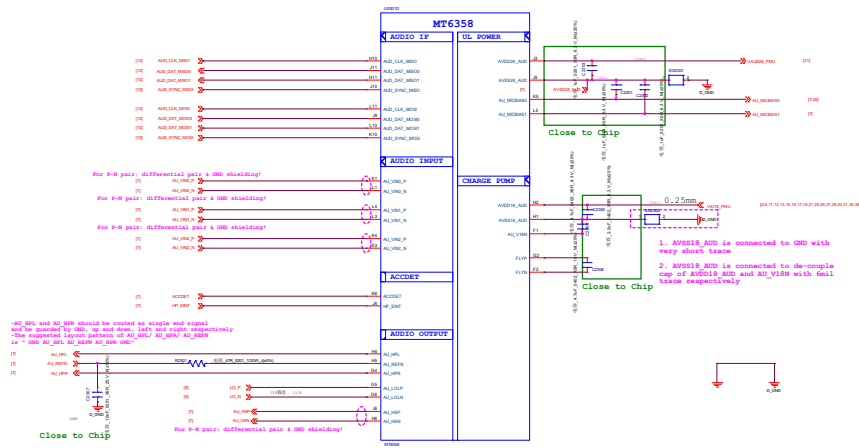
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 19_SENSORS_ALS_PS_A		
Size D	Project AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024	Sheet 19 of 36	



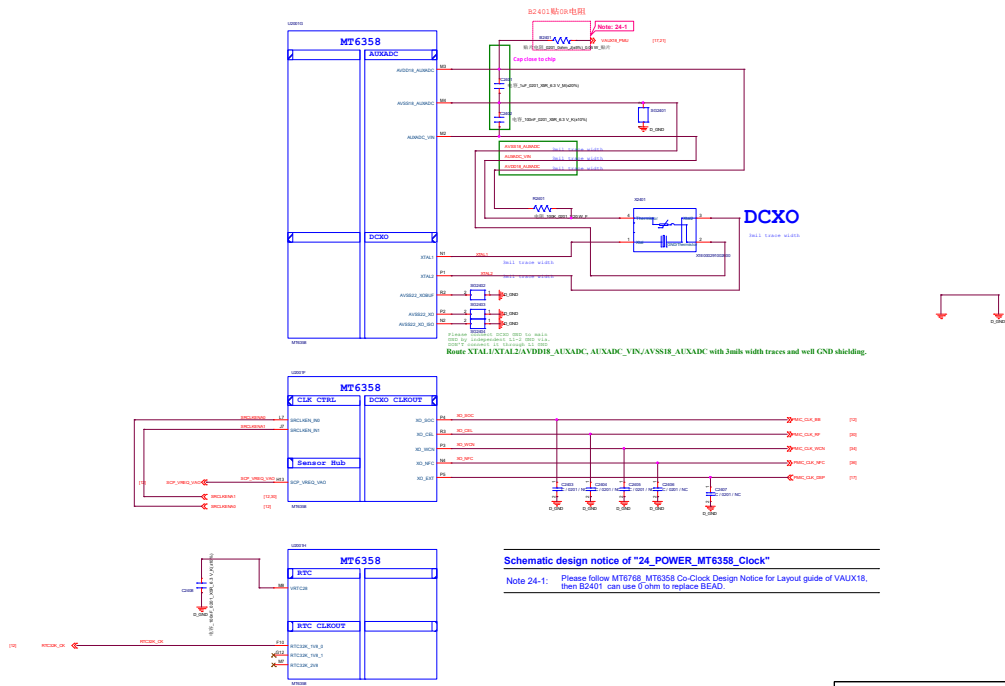
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 20_POWER_MT6358-Buck			
Size E	Project	AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024		Sheet 20 of 40	



华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 22_POWER_MT6358-IF		
Size D	Project AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024	Sheet 22 of 36	

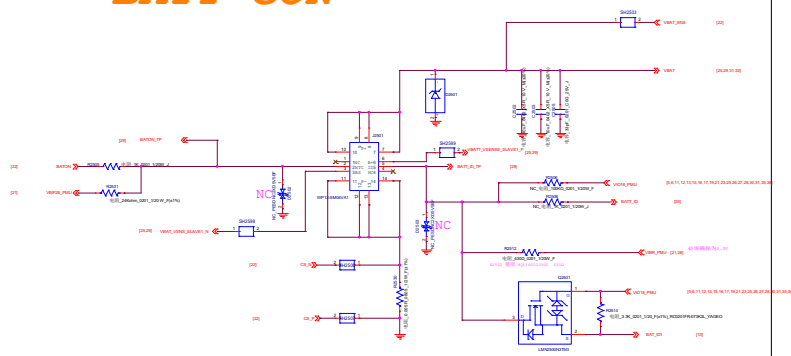


华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title 23_POWER_MT6358-Audio			
Size D	Project	AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024		Sheet 23 of 36	

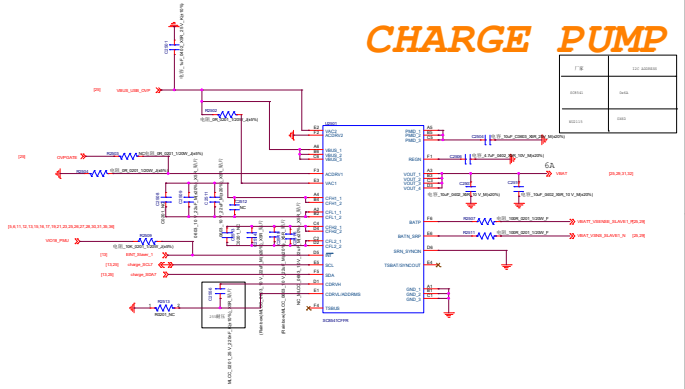


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		24_POWER_MT6358_Clock	
Size	Project	AL7015A_MB_V1	Rev
D			V1
Date:	Monday, January 01, 2024	Sheet	24 of 36

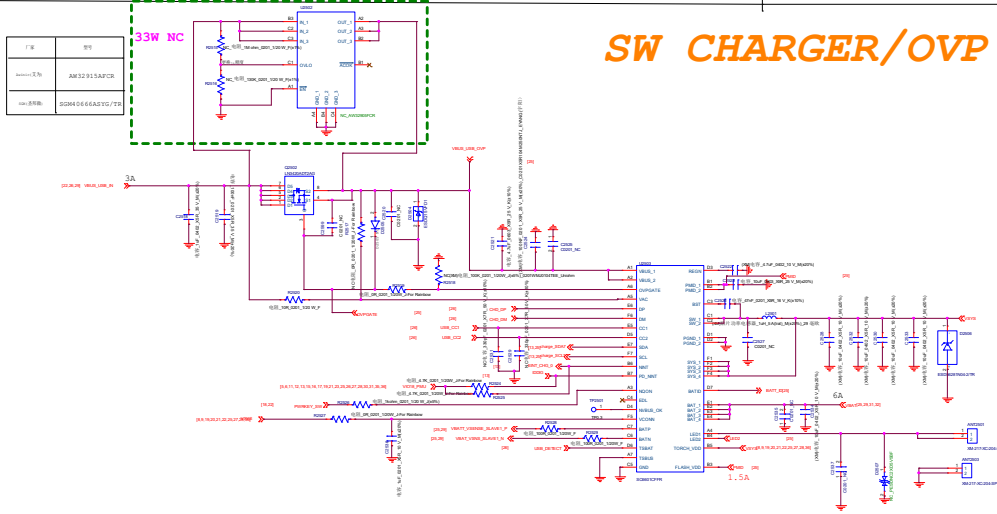
BATT CON



CHARGE PUMP



SW CHARGER/OVP

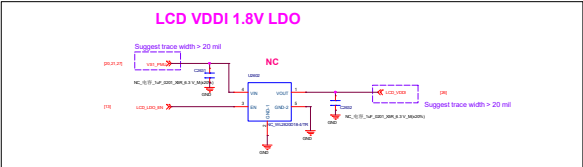
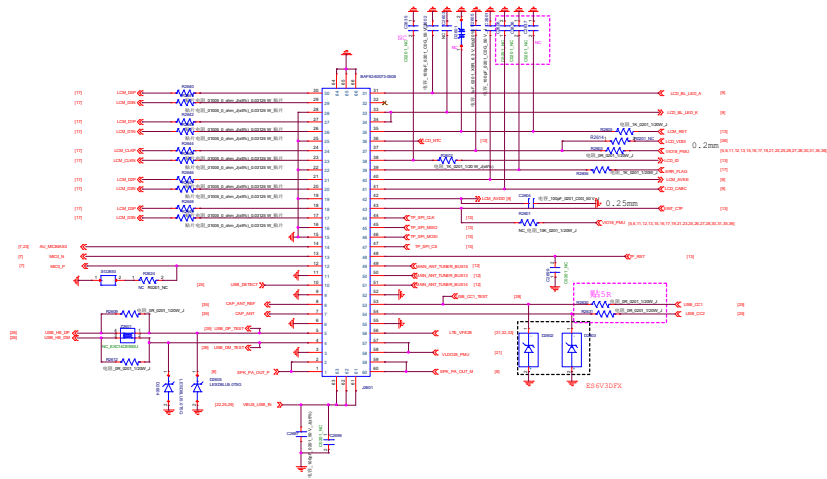


闪光灯TVS NC

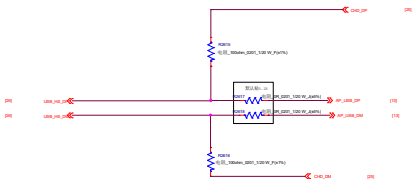


华勤通讯 Huaqin Telecom Technology Com., Ltd			
Title 26_POWER_SubPMIC-Charger+CONN			
Size D	Project AL7015A_MB_V1	Rev v1	
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LINK



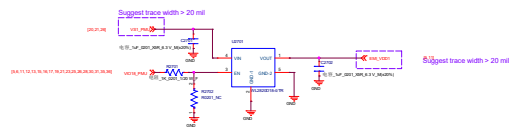
DP/DM



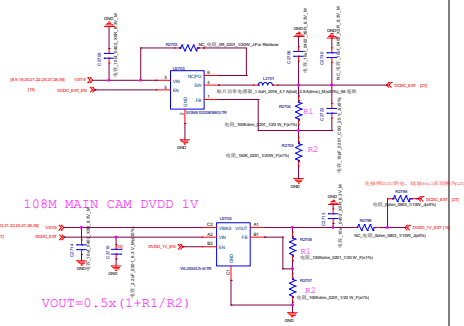
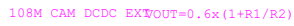
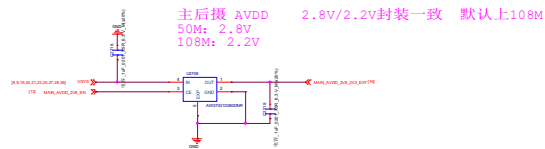
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 26_LCD 60PIN			
Size D	Project	AL7015A_MB_V1	Rev V1
Date: Thursday, February 01, 2024		Sheet 26 of 36	



Note: 28-5

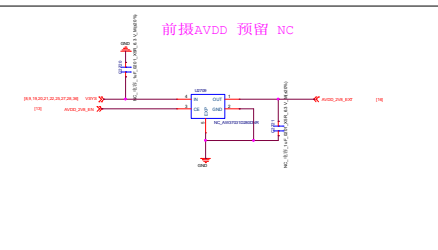


 Suggest trace width > 20 mil


$$V_{OUT} = 0.5 \times (1 + R_1/R_2)$$


主后撮 AVDE
50M: 2.8V
108M: 2.2V

108M: 2.2V



Schematic design notice of "28_POWER_ThirdParty-Power"

Note 28-1: VA12 Layout placement please close to AP

Note 28-2: VS2 Buck Layout placement please close to PMIC MT6357

Note 28-3: VCN33 LDO Layout placement please close to MT6631

华勤通讯 Huaqin Telecom Technology Com.,Ltd

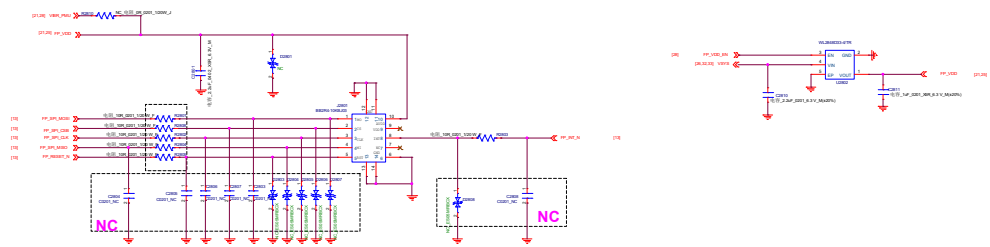
Title	27_POWER_ThirdParty_Powers
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Size D	Project AL7015A_MB_V1
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Rev
v1

Date: Monday, January 01, 2024 Sheet 27 of 36

Finger Print



华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 28_Fingerprint		
Size D	Project AL7015A_MB_V1	Rev v1
Date: Monday, January 01, 2024		Sheet 28 of 36

工厂测点



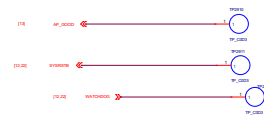
VBAT



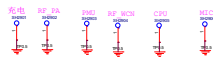
串口



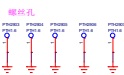
研发测点



屏蔽罩



光学定位孔

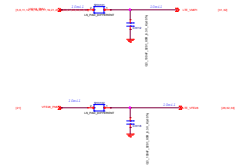
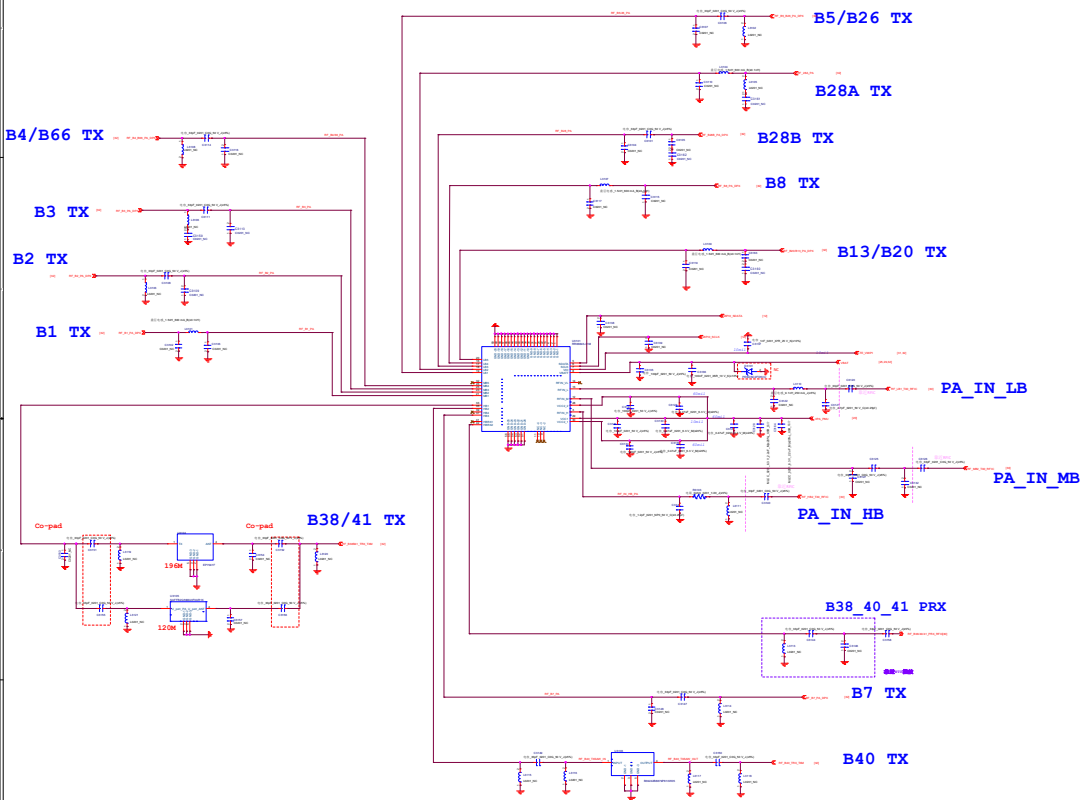


提升



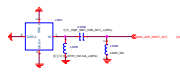
华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title 29_DEBUG_IO		
Size D	Project AL7015A_MB_V1	Rev V1
Date: Monday, February 05, 2024	Sheet 29 of	36

RF Power Supply



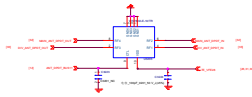
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title AL7015A_RF_MT6177_RF_TX			
Size	Project	AL7015A_MB_V1	Rev v1
Custom	Custom		
Date:	Monday, January 01, 2024	Sheet	31 of 36

MAIN_ANT

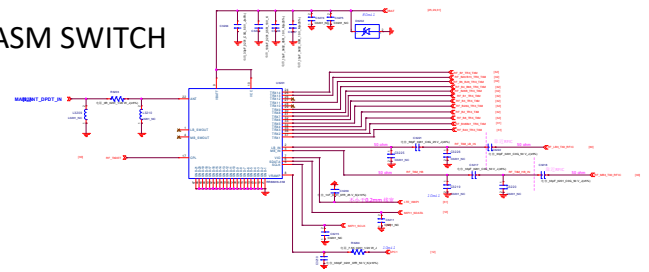


DPDT SWITCH

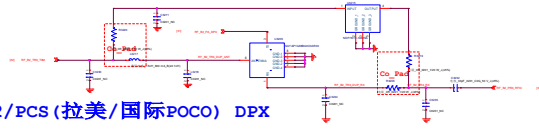
天线接口使用继电器，请按照规格，不要有错接



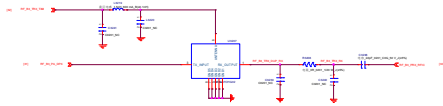
ASM SWITCH



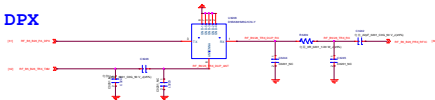
B2/PCS (拉美/国际POCO) DPX



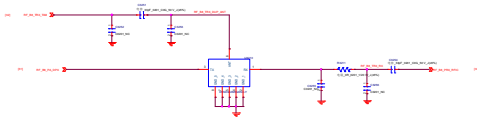
B3 DPX



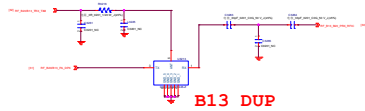
B5/B26 DPX



B8 DPX

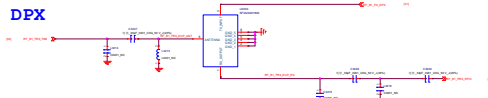


B13/20 DPX

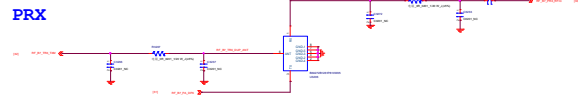


B13 DUP

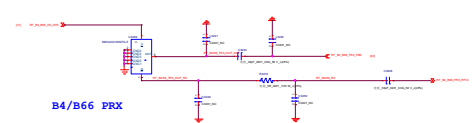
B1 DPX



B7 PRX

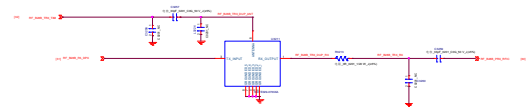


B4/B66 DPX

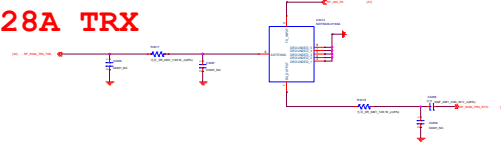


B4/B66 PRX

B28B TRX

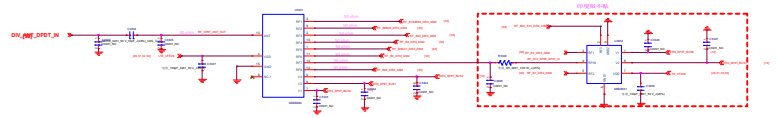


B28A TRX

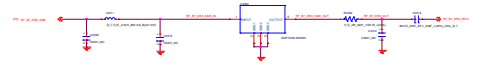


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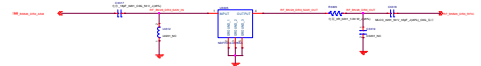
DRX Switch



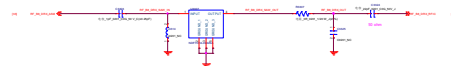
B7 DRX



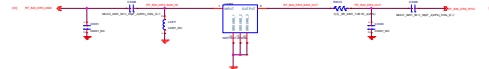
B5/B26 DRX



B8 DRX



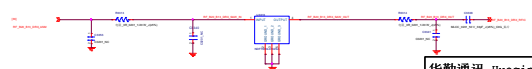
B40 DRX



B38/41 DRX

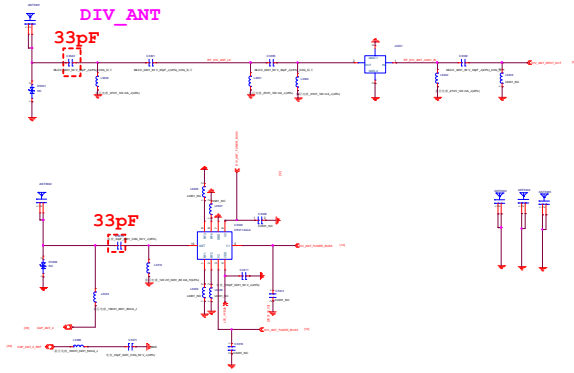


B20/13 DRX

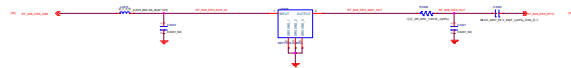


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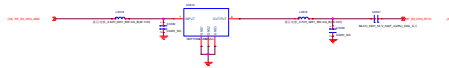
B28 DRX



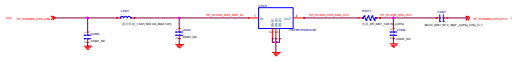
B2 DRX

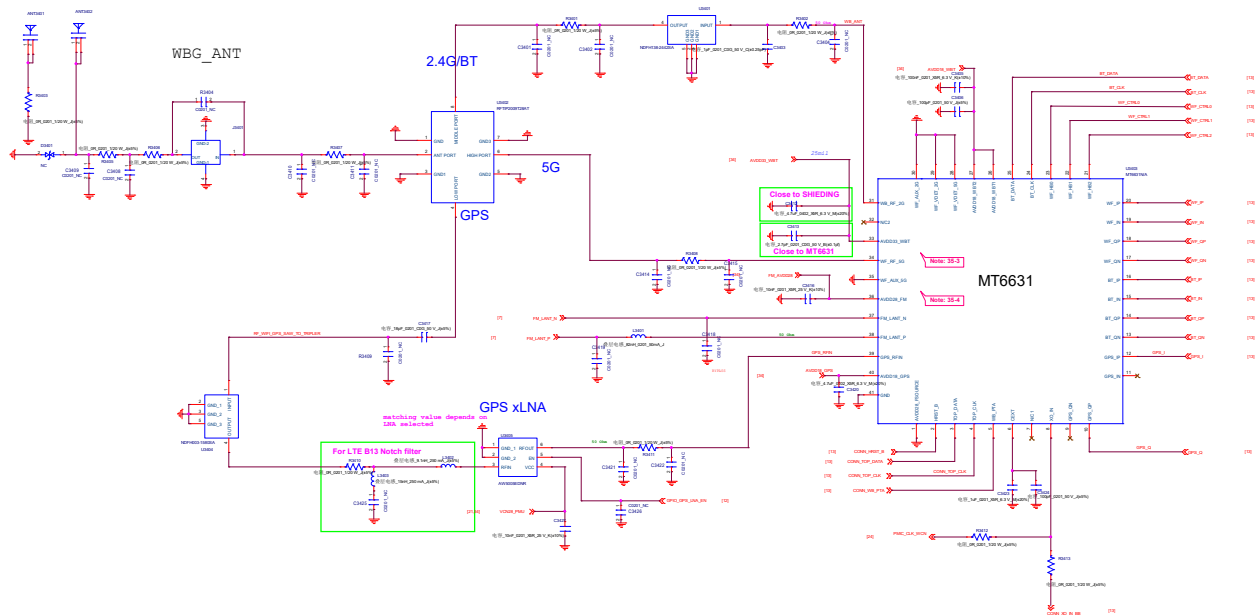


B3 DRX

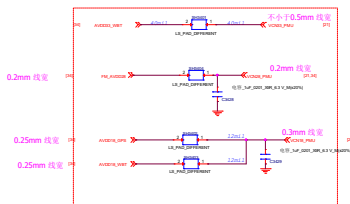


B1/B4/B66 DRX





WCN Power Supply



Schematic design notice of "S1_CONNECTIVITY_CONSYS_MT6631"

Note 35-1: For R5015 size, please select 0402 size or larger one

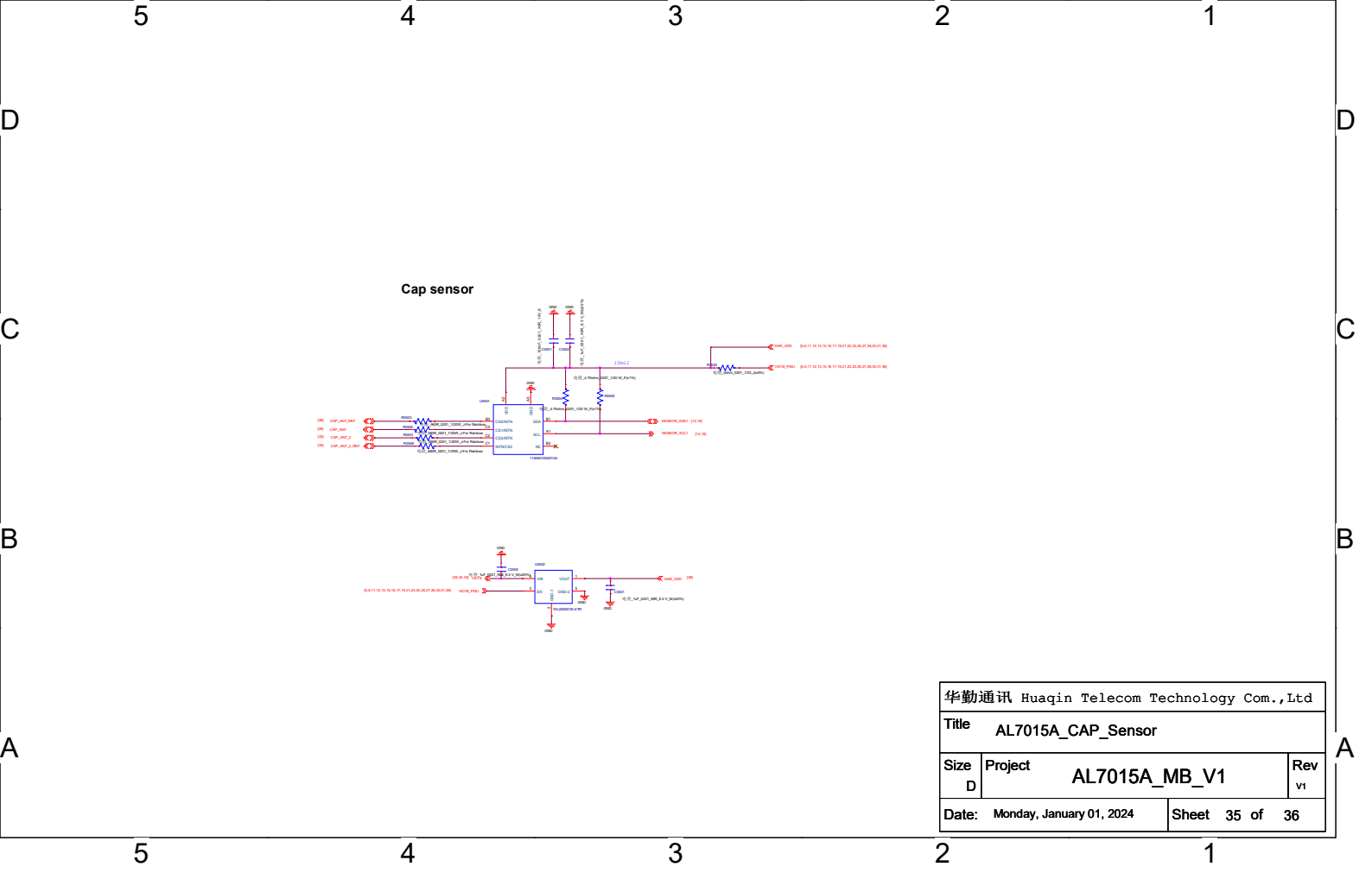
Note 35-2: Please refer to MT6765 Baseband design notice for VCN33 LDO selection guide

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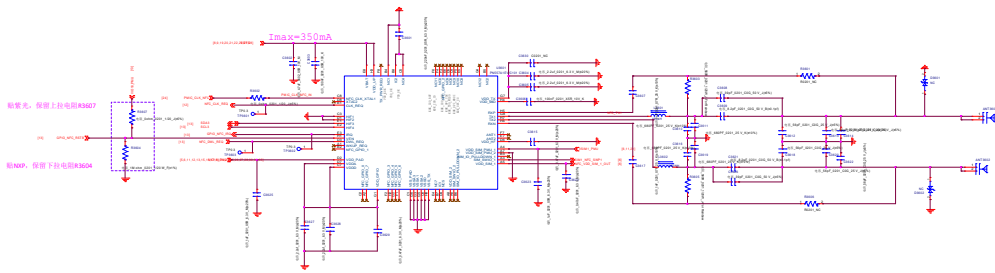
Title AL7015A_CONNECTIVITY_CONSYS_MT6631

Size D Project AL7015A_MB_V1 Rev V1

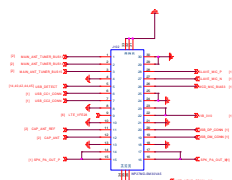
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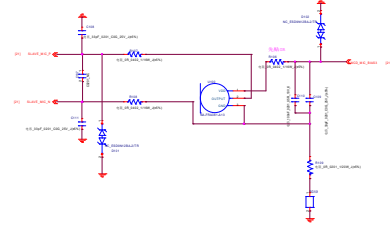
华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title AL7015A_CAP_Sensor		
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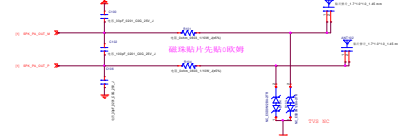


MAIN MIC

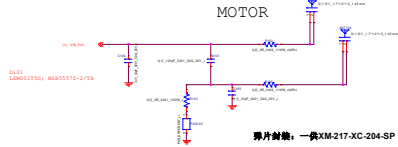


SPK

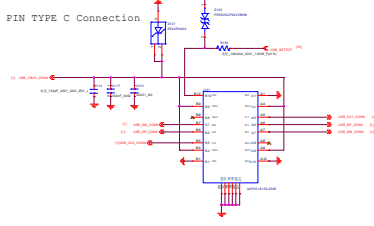
扬声器：一供XM-217-XC-204-SP



MOTOR



16 PIN TYPE C Connection



光学定位孔

